Tusher Bhomik

+8801690090528 | tusherbhomik@gmail.com | Palashi, Newmarket, Dhaka-1205

Portfolio | LinkedIn | GitHub | LeetCode | Kaggle

Education

- **Bangladesh University of Engineering and Technology** — *2022 – Present (currently at 8th semester)*
  Bachelor of Science in CSE — Dhaka, Bangladesh

Projects

- **HealthSync** — *2025*
  Spring Boot, React, PostgreSQL, Firebase — GitHub
  - Developed a comprehensive healthcare platform connecting patients and doctors with appointment scheduling, digital prescription management, and medical history tracking.
- **Arxiv Research Paper RAG System** — *2026*
  Python, SQL, SQLite, FastAPI, ChromaDB, Data Analysis — GitHub
  - Built an end-to-end **data pipeline** for ingesting, cleaning, and analyzing arXiv research paper data using Python and SQL.
  - Developed a **RAG-based API system** using FastAPI and ChromaDB for semantic retrieval and question answering.
  - Generated analytical insights with **visualizations** and implemented automated benchmarking using query-based evaluation.
- **Mini C Compiler** — *2024*
  CSE-310 Compiler Design, C, 8086 Assembly — GitHub
  - Built a **mini C compiler** from scratch in four phases: Symbol Table, Lexical Analysis, Semantic Analysis, and Intermediate Code Generation.
  - Generated **8086 assembly** as intermediate output to validate end-to-end compilation workflow.
- **Enhanced xv6-riscv Kernel** — *2024*
  C, Assembly, Makefile — GitHub
  - Implemented an **MLFQ scheduler** with aging and added **kernel-level threading** support.
  - Added custom system calls (`trace`, `sysinfo`) and a Bash autograder for testing.
- **Dealytics** — *2025*
  Marketplace Analytics Platform — GitHub
  - Building Bangladesh's first **analytics-driven marketplace aggregator** to track real-time price trends and predict discounts across markets.
  - Implementing **data-driven inventory optimization** for sellers and a secure negotiation system to facilitate transparent trading.

Research

- **Fall Risk Estimation from Gait using Deep Learning (Ongoing Thesis)** — *2025 – Present*
  Supervisor: Mahmuda Naznin, Dept. of CSE, BUET — Dhaka, Bangladesh
  - Developing an interpretable deep learning framework to estimate fall vulnerability from motion capture gait data by detecting joint-level deviations and inferring biomechanical causes of fall risk.
- **Improving Function Calling in Large Language Models using Reinforcement Learning (Ongoing)** — *2026 – Present*
  Supervisor: Md. Rizwan Parvez, QCRI Research Institute — Doha, Qatar
  - Investigating reinforcement learning-based optimization to enhance the reliability and accuracy of function calling behavior in large language models for complex tool-use scenarios.

Achievements

- **5th Place — Tech Career Hunt 2.0**, organized by Shohoj Coding.

Skills

- **Programming:** Python, C++, C, Java, JavaScript | **Databases:** MySQL, PostgreSQL, MongoDB
- **ML/AI:** PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, OpenCV, LLMs, vLLM
- **Tools:** Git, Docker, Linux | **Frontend:** React, Next.js, Material-UI

Relevant Coursework

- **Relevant Coursework** Data Structure, Algorithm, Computer Networks, Database Management System, Compiler Design, Operating System, Software Engineering, Artificial Intelligence, Machine Learning, Computer Architecture, Theory of Computation, Computer Graphics

Additional Information

Languages: English, Bengali. **Interests:** Photography, Traveling, Badminton, Cycling.